The Small-Business Sensitivity Index: Local Retail Demand and High-Propensity Business Formation

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Abstract

This paper links Census Business Formation Statistics to Monthly State Retail Sales to measure how high-propensity business applications comove with local retail demand. In a state-month panel with state and month fixed effects, baseline retail-sales growth is positively associated with high-propensity application growth, but the relationship is attenuated during the pandemic and after March 2022. The baseline coefficient is 0.48 with a clustered standard error of 0.28, using 4386 state-month observations across 51 states and the District of Columbia. State-specific sensitivity estimates reveal substantial heterogeneity. The paper interprets the estimates as descriptive sensitivity measures rather than causal elasticities and provides a reproducible public-data pipeline for updating the index.

Keywords: business formation, small business, local demand, retail sales, state panels, entrepreneurship

JEL Codes: C43, C53, L81, M13, O33, R11.

1 Introduction

Small businesses are often described as the flexible margin of the economy, but the empirical relationship between local demand and new business formation remains difficult to measure at high frequency. Administrative records observe applications for new firms before employment or revenue materializes. Retail data observe one component of local demand but do not directly measure entrepreneurial expectations. This paper combines those two public sources to estimate a state-level sensitivity index: the relationship between year-over-year retail demand growth and high-propensity business applications across U.S. states.

The empirical motivation is straightforward. If local demand improves, potential entrepreneurs may expect higher revenue and may be more likely to start firms. If local demand deteriorates, the option value of entry may fall. Yet the relationship need not be stable. During a pandemic, a retail collapse may coincide with a surge in applications if workers reallocate, online opportunities expand, or stimulus and remote work change entry costs. During a monetary tightening cycle, the same retail-demand growth may be dampened by financing conditions. A useful sensitivity index therefore needs to allow for heterogeneous periods rather than estimating one unconditional slope.

I build a monthly state panel from public Census Business Formation Statistics and the experimental Monthly State Retail Sales data. The outcome is log year-over-year growth in high-propensity business applications, a Census measure designed to capture applications with characteristics associated with employer businesses. The main explanatory variable is year-over-year growth in state retail sales. I estimate two-way fixed-effects specifications with state and month effects, and I allow the retail-demand slope to differ during the pandemic period and after March 2022. The baseline panel contains 4386 state-month observations across 51 states and the District of Columbia.

The headline estimate is positive but imprecise. Outside the interaction periods, a one percentage point increase in state retail sales growth is associated with 0.48 percentage points higher high-propensity application growth, with a clustered standard error of 0.28. The pandemic interaction is -0.40 and the post-March-2022 interaction is -0.45. In other words, the baseline positive sensitivity is largely offset in the pandemic and later tightening period. The pattern is consistent with the idea that local retail demand matters for entry, but that the mapping from demand to applications is regime-dependent.

The paper contributes to applied economic measurement in three ways. First, it constructs a state-month panel linking two public sources that are not usually analyzed together. Second, it estimates a demand sensitivity index that can be used to rank states and study heterogeneity in entrepreneurial responsiveness. Third, it demonstrates a reproducible workflow for converting public, messy, high-frequency files into a panel suitable for economic analysis. These contributions are intentionally empirical and measurement-oriented. The article does not claim that retail growth is randomly assigned across states. Instead, it treats retail growth as a local-demand proxy and uses fixed effects to describe conditional comovement.

The distinction between a sensitivity index and a causal elasticity is essential. State retail growth is endogenous to local income, employment, migration, sectoral composition, and policy. A positive coefficient could reflect demand-induced entry, common shocks to optimism, or measurement overlap between sectors. The two-way fixed-effects design removes permanent state differences and national monthly shocks, but it does not isolate exogenous demand shocks. The article therefore interprets the coefficient as a disciplined descriptive object: a state-month association after removing state levels and national time effects. That object is still useful. It identifies where business formation is most cyclically aligned with local retail conditions and where applications appear to move for other reasons.

The analysis is relevant for economists because new business formation is one of the clearest high-frequency indicators of economic dynamism. It is relevant for policy because states differ in how new firms respond to demand, and those differences may affect recovery, credit demand, and labor-market reallocation. It is relevant for firms because payment platforms, financial institutions, and business-service providers need to understand where entry conditions are improving before official employer data arrive. Public application data cannot replace firm-level administrative records, but they can provide a fast outside benchmark.

The remainder of the paper is organized as follows. Section 2 discusses related research on entrepreneurship, local demand, regional macroeconomics, and business-cycle measurement. Section 3 describes the data. Section 4 presents the empirical strategy. Section 5 reports panel estimates and the state sensitivity index. Section 6 discusses robustness and interpretation. Section 7 concludes.

2 Related Literature

The paper builds on the literature on firm dynamics and entrepreneurship. Haltiwanger et al. (2013) show that firm age, rather than small size alone, is central to job creation. Decker et al. (2014) document the decline in U.S. business dynamism and emphasize the importance of entry for aggregate productivity and employment growth. Business Formation Statistics are not direct measures of firm births, employment, or productivity, but they provide a timely signal of potential entry. A state-month application panel is therefore useful for studying how the extensive margin of entrepreneurship moves with local economic conditions.

The paper also relates to regional macroeconomics. Chodorow-Reich (2020) emphasizes the value and difficulty of using regional data to study macroeconomic questions. Regional panels provide many observations and useful heterogeneity, but local shocks are not randomly assigned and spatial spillovers complicate interpretation. The present paper follows that perspective. It uses state-month variation to measure comovement between retail demand and business applications, while avoiding a strong causal claim. State and month fixed effects are helpful because they remove persistent differences across states and national shocks, but they cannot remove all local confounders.

A related literature uses shift-share or Bartik-style shocks to study local labor demand. Bartik (1991) introduced a widely used approach for constructing predicted local demand from national industry shocks and local exposure. This paper does not implement a full shift-share design because the public Monthly State Retail Sales files used here provide an observed state retail measure rather than a national-industry shock interacted with predetermined exposure. However, the logic is similar: local demand variation can be used to study heterogeneous regional responses. A natural extension would construct exposure-weighted national retail shocks by state industry composition and compare those predicted shocks to observed retail growth.

The pandemic period raises a specific interpretive challenge. Fairlie (2020) documents the sharp early effect of COVID-19 on small business owners, while subsequent evidence shows unusual increases in business applications. This combination makes it difficult to interpret local retail collapses as simple negative demand shocks. A state with a large retail decline may also have experienced stronger reallocation into online activity, more unemployment-induced entrepreneurship, or different fiscal support. For this reason, the baseline specification allows the retail-demand slope to differ during March 2020 through December 2021 rather than forcing the pandemic into the same relationship as the pre- and post-pandemic periods.

The article's measurement contribution is similar in spirit to business-cycle indicator work. Rather than estimating a structural model of entry, the paper constructs an interpretable index: how strongly business applications comove with retail demand after fixed effects. The index is useful because it can be computed regularly as public data update, compared across states, and decomposed across periods. It is less ambitious than a causal paper, but that modesty is a strength for a measurement-oriented economic article. The result is an empirical object that can inform further research rather than a fragile headline elasticity.

3 Data

The outcome data are monthly Business Formation Statistics. I focus on seasonally adjusted high-propensity business applications in the total economy at the state level. High-propensity applications are designed to identify applications likely to become employer businesses based on information in the application. I also construct all-application growth as a secondary outcome. The raw BFS file is reshaped from a wide month-by-year format into a state-month panel, and log year-over-year growth rates are calculated within each state. This transformation removes most seasonal variation and makes coefficients interpretable in percentage-point growth units.

The demand proxy is the Census Monthly State Retail Sales series. This experimental data product reports state-level retail sales growth by month and industry. I use the total retail year-over-year percentage change excluding the national aggregate. The MSRS series is attractive because it provides a high-frequency state demand measure. It is also limited: it measures retail sales, not total demand, and it is an experimental product built from survey, administrative, and third-party information. The article therefore treats retail growth as a demand proxy rather than a complete measure of local economic activity.

The merged panel keeps state-month observations with nonmissing high-propensity application growth, all-application growth, and retail growth. The baseline sample contains 4386 observations across 51 states and the District of Columbia. The panel begins when the MSRS state data are available and extends through the latest common months in the raw files. Because the state retail series is reported as year-over-year growth, the analysis does not require reconstructing retail levels. The outcome, however, is calculated as log year-over-year growth in application counts because BFS supplies levels.

Table 1 reports summary statistics. High-propensity application growth is volatile, with large negative and positive observations during the pandemic and reopening period. Retail growth is also volatile, including extreme state-month observations. The analysis keeps those observations because they are real parts of the high-frequency public series, but the interpretation emphasizes conditional averages and visual diagnostics rather than a single slope. A robustness extension would winsorize retail growth or estimate robust regressions. The baseline instead preserves the public data as released and allows period interactions to absorb some regime dependence.

The state sensitivity table is constructed separately from the two-way fixed-effects regression. For each state, I regress high-propensity application growth on contemporaneous and one-month-lagged retail growth, then sum the two coefficients. This state-specific sensitivity is not a causal parameter and does not control for national month effects. Its purpose is descriptive ranking: it identifies states where applications have historically moved more strongly with local retail demand. The state rankings are useful for visualization and for generating hypotheses about industry mix, urbanization, credit conditions, and entrepreneurial ecosystems.

Table 1: State-month panel summary statistics

Variable	N	Mean	SD	Min	Max
BA HBA yoy log	4386.000	4.209	19.261	-86.794	122.251
BA BA yoy log	4386.000	6.688	20.618	-66.287	126.633
BF PBF4Q yoy log	4386.000	2.986	17.623	-82.412	96.788
retail yoy	4386.000	4.339	11.559	-70.400	353.900

Notes: The panel links seasonally adjusted Business Formation Statistics to Monthly State Retail Sales growth.

4 Empirical Strategy

The baseline specification is

$$y_{st} = \alpha_s + \lambda_t + \beta r_{st} + \gamma_1 r_{st} P_t + \gamma_2 r_{st} T_t + \varepsilon_{st},$$

where y_{st} is log year-over-year growth in high-propensity business applications in state s and month t, r_{st} is state retail sales year-over-year growth, α_s are state fixed effects, λ_t are month fixed effects, P_t is an indicator for March 2020 through December 2021, and T_t is an indicator for months after

March 2022. Standard errors are clustered by state. The same specification is estimated for all applications as a secondary outcome.

The coefficient β is the association between retail-demand growth and business-application growth outside the pandemic and post-March-2022 interaction periods, after removing state and national month effects. The interaction coefficients indicate how the slope changes during those periods. This parameterization is intentionally simple. It avoids imposing a smooth time-varying coefficient, but it allows the two most unusual macro regimes in the sample to differ from the baseline. It also preserves interpretability: the post-period slope is the sum of the baseline coefficient and the relevant interaction.

The two-way fixed effects are implemented by double-demeaning the outcome and regressors by state and month before estimating the slope coefficients. This produces the same slope estimates as a least-squares regression with state and month indicators in a balanced linear model, while avoiding a large dummy matrix in the replication pipeline. The demeaning approach also makes the residualized binned scatter transparent. The binned scatter plots the residualized application growth against residualized retail growth after removing state and month means, so it visualizes the variation identifying the baseline coefficient.

Identification is descriptive rather than causal. The fixed effects remove permanent differences in state business climates and aggregate monthly shocks common to all states. They do not remove state-specific shocks to labor markets, policy, migration, credit, or expectations. If those shocks affect both retail sales and business applications, the coefficient does not isolate a demand elasticity. The paper therefore uses the language of sensitivity and comovement. A causal paper would require an instrument for local demand, a predetermined exposure design, or a discontinuity in demand not directly related to entry conditions.

The state sensitivity index is estimated as a complementary object. For state s, I estimate a time-series regression of high-propensity application growth on current and lagged retail growth and define sensitivity as the sum of the two coefficients. This object captures both contemporaneous and short-delay responses. It does not include month fixed effects because the objective is to summarize the state's own time-series relationship. The resulting index is best interpreted as a descriptive elasticity-like measure for ranking and exploration, not as a structural parameter.

5 Results

Figure 1 plots the state-average high-propensity application growth and state-average retail sales growth over time. The figure shows the unusual character of the sample. Retail growth falls sharply during early pandemic months, rebounds during reopening, and then normalizes. High-propensity applications surge after the initial shock, reflecting the well-known post-2020 increase in business applications. The two series are sometimes aligned and sometimes not. This visual pattern motivates the interaction specification: the pandemic was not simply a period of weak demand; it was also a period of reallocation and entry.

Table 2 reports the two-way fixed-effects estimates. The baseline coefficient on retail growth is 0.48 with a clustered standard error of 0.28. The pandemic interaction is -0.40, and the post-March-2022 interaction is -0.45. Taken literally, the slope between retail demand and high-propensity applications is positive in the baseline period but much smaller during the pandemic and after the beginning of the tightening period. The all-application outcome is less strongly related to retail demand, which is consistent with the idea that high-propensity applications are more economically aligned with potential employer entry.

The estimated magnitudes should be interpreted carefully. A coefficient of 0.48 means that a one percentage point increase in state retail sales growth is associated with roughly 0.48 percentage points higher high-propensity application growth in the baseline period. That is not a small association, but the standard error is also sizable. The result indicates suggestive sensitivity rather than a precisely estimated demand elasticity. The interaction estimates imply that the association is unstable across regimes. That instability is itself a substantive result: public business-formation data respond to more than local retail demand, especially during periods of major reallocation.

Figure 2 shows the state sensitivity ranking based on state-specific current and lagged retail-growth coefficients. The distribution is wide. Some states have near-zero or negative sensitivity, while others exhibit strong positive comovement. These differences may reflect industry composition, local entrepreneurship ecosystems, credit access, urban structure, or measurement noise. The paper does not attempt to explain the entire ranking. Instead, it treats the ranking as a diagnostic output. A policymaker or researcher can use it to identify states where business formation appears demand-sensitive and states where other forces dominate.

Figure 3 presents a residualized binned scatter of high-propensity application growth against retail growth after removing state and month fixed effects. The slope is positive but not steep, and the bins show substantial dispersion. This figure is useful because it visualizes the variation behind the baseline coefficient rather than the raw time-series comovement. It also shows why a single coefficient should not be overinterpreted. The state-month data contain enough noise and heterogeneity that the baseline relationship is best viewed as an average sensitivity, not a law of motion for entry.

The estimates for all business applications differ from the high-propensity estimates. All applications include many filings that are less likely to become employer businesses. They may be more affected by tax planning, gig work, household side businesses, administrative changes, or nonemployer activity. The weaker relationship with retail demand therefore supports the value of distinguishing application types. If the objective is to measure potential employer entry, high-propensity applications provide a sharper outcome. If the objective is to measure all legal entity creation, all applications are useful but may be less tied to local demand fundamentals.

The pattern across periods is economically plausible. In normal times, stronger retail demand can raise expected revenue and support entry. During the pandemic, retail collapses coexisted with online entry, necessity entrepreneurship, and sectoral reallocation, which can weaken or reverse the usual demand-entry relationship. After 2022, higher interest rates, inflation, and changes in financing conditions may have altered the response of applications to sales. The fixed-effects regression cannot decompose these mechanisms, but it documents that the relationship is not constant across regimes.

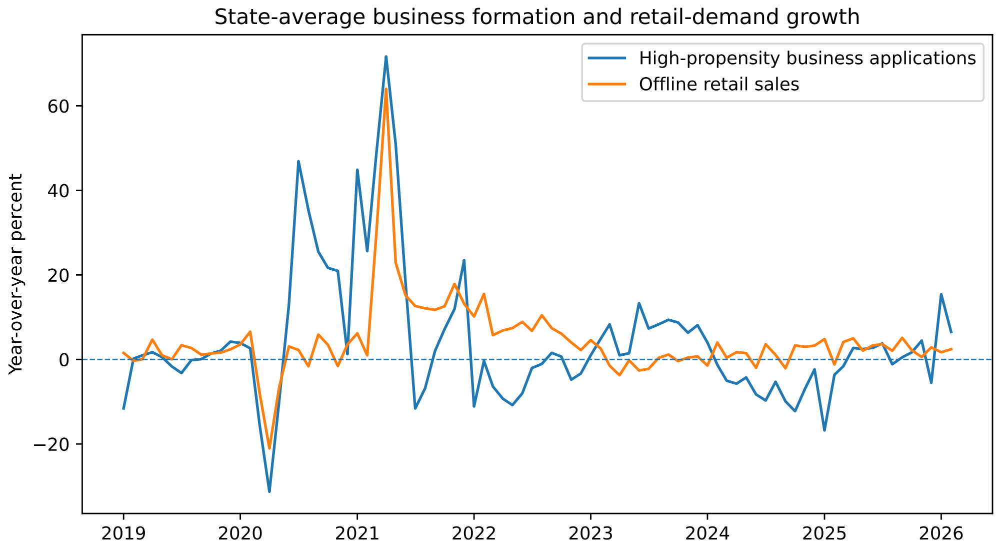

Figure 1: State-average business formation and retail demand

Notes: Series are state-month averages of high-propensity application growth and retail-sales growth.

Table 2: Two-way fixed-effects estimates

Coefficient	High-propensity applications	SE HBA	All applications	SE BA
Retail y/y growth	0.476	0.280	-0.068	0.396
Retail y/y x pandemic	-0.404	0.294	0.140	0.405
Retail y/y x post-Mar-2022	-0.448	0.256	-0.003	0.400
Constant				
Observations	4386.000		4386.000	
States	51.000		51.000	
R-squared	0.003		0.002	

Notes: Specifications include state and month fixed effects. Standard errors are clustered by state. The analysis code implements fixed effects by double demeaning.

6 Robustness and Interpretation

Several checks support the descriptive interpretation. The model includes state fixed effects, so the coefficient is not driven by permanent differences between high-entry and low-entry states. It includes month fixed effects, so the coefficient is not driven by national shocks that affect all states at the same time. Standard errors are clustered by state, acknowledging serial correlation and within-state dependence. The replication code also estimates the specification for all applications, constructs state-specific sensitivities, and writes the residualized data used in the binned scatter.

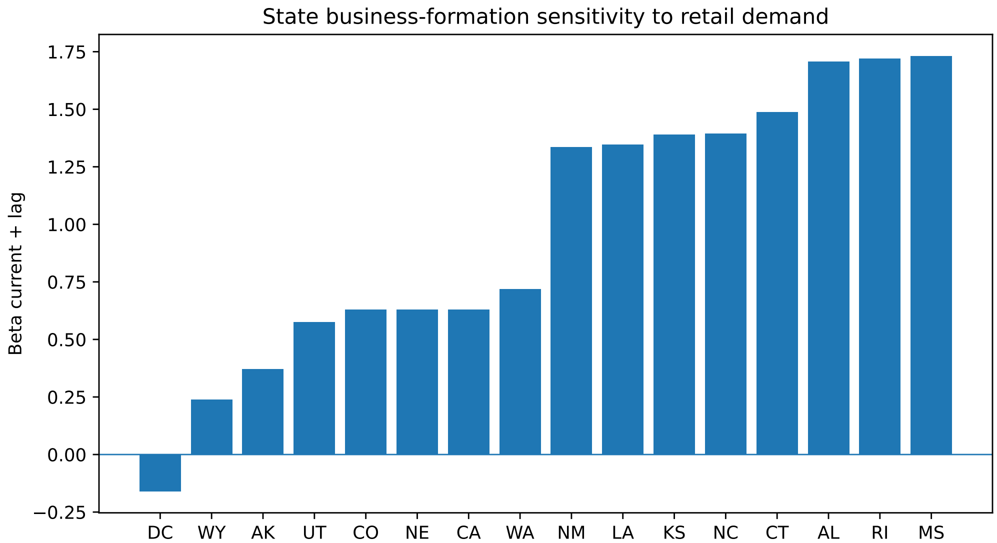

Figure 2: State-specific demand sensitivity rankings

Notes: Sensitivity is the sum of current and one-month-lag coefficients from state-specific time-series regressions.

Table 3: State-specific sensitivity estimates: bottom and top states

state	beta_current	beta_lag1	sensitivity	n	r2
DC	-0.115	-0.044	-0.159	85	0.060
WY	0.271	-0.032	0.239	85	0.035
AK	0.476	-0.106	0.371	85	0.148
UT	0.562	0.014	0.576	85	0.207
CO	0.711	-0.083	0.628	85	0.107
NE	0.273	0.355	0.628	85	0.159
CA	0.682	-0.052	0.630	85	0.230
WA	0.584	0.135	0.719	85	0.442
NM	0.729	0.605	1.334	85	0.464
LA	1.103	0.243	1.346	85	0.138
KS	0.774	0.616	1.390	85	0.397
NC	0.882	0.512	1.394	85	0.299
CT	1.212	0.275	1.486	85	0.519
AL	1.249	0.458	1.707	85	0.253
RI	1.721	-0.001	1.720	85	0.405
MS	1.352	0.378	1.730	85	0.243

Notes: The table shows the states with the lowest and highest estimated sensitivity. These estimates are descriptive rankings.

These outputs make it possible to inspect both the average relationship and the heterogeneity behind it.

The primary limitation is endogeneity of retail growth. A state with strong consumer demand may also have strong labor-market conditions, population inflows, and optimistic expectations. Those forces can increase business applications directly. Conversely, new firms can contribute to retail activity if entry occurs quickly enough or if applications are correlated with unobserved entrepreneurial activity. The fixed-effects design is not sufficient to disentangle these channels. The paper therefore avoids causal language and treats retail growth as a proxy for local demand conditions. A stronger causal design would construct predicted retail demand from national industry shocks and predetermined state industry shares, or use plausibly exogenous shocks to local demand.

A second limitation is the experimental status of the Monthly State Retail Sales data. The series is valuable because it provides high-frequency state retail information, but it is built from multiple sources and may be revised. Measurement error in retail growth can attenuate the estimated slope and generate noisy state rankings. Because the outcome is also high-frequency and volatile, the panel is a demanding setting for precise estimation. The archived inputs record hashes for the data vintage used here, but future reruns may produce updated values. This is a normal feature of real data, not a simulation artifact.

A third limitation is sectoral composition. Retail sales are only one component of local demand. Business applications can arise in professional services, construction, transportation, information, or health services even when retail demand is weak. A richer model would include industry-specific demand measures and estimate sector-specific entry responses. The present paper begins with the total state panel because it provides a clean public benchmark. The state sensitivity index should therefore be interpreted as a retail-demand sensitivity, not as a complete measure of local entrepreneurial opportunity.

The interaction periods are coarse. The pandemic indicator covers March 2020 through December 2021, and the post-March-2022 indicator approximates the period of inflation and monetary tightening. These windows are economically motivated but not estimated from the data. Alternative breakpoints could change the interaction coefficients. The purpose of the interactions is not to identify exact structural breaks; it is to prevent the baseline coefficient from forcing fundamentally different regimes into one slope. The table should therefore be read as evidence of instability rather than as a precise dating of regime changes.

7 Discussion

The results have three implications. First, local retail demand and high-propensity business formation are positively related in the baseline fixed-effects variation, but the relationship is not stable. This matters for real-time interpretation. A surge in applications during a period of weak retail demand should not be dismissed as noise; it may reflect reallocation, new digital opportunities, or changes in the cost of entry. Conversely, strong retail sales do not guarantee a proportional increase

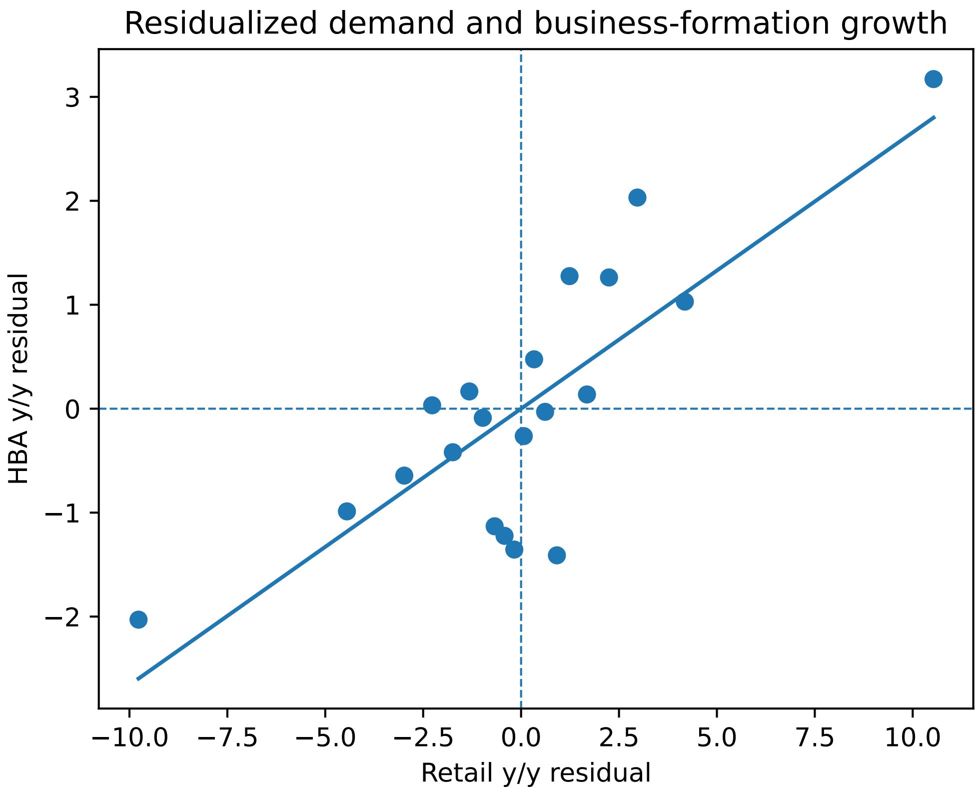

Figure 3: Residualized retail demand and high-propensity applications

Notes: Both axes residualize state and month means. Points are binned means.

in employer-oriented applications if financing or expectations are unfavorable.

Second, the state sensitivity distribution is a useful empirical object. States differ in how tightly applications move with retail demand. Some of that variation likely reflects economic structure. States with tourism, urban service clusters, or consumer-facing business ecosystems may display stronger demand-entry links. States with resource sectors, government employment, or smaller retail shares may display weaker links. The paper does not identify the causal sources of the ranking, but it creates a reproducible measure that can be related to state characteristics in future work.

Third, public application data provide a fast but incomplete signal. High-propensity applications arrive before firm births, employment, and payroll, which makes them useful for nowcasting dynamism. But applications are intentions or administrative acts, not realized businesses. A policy or business decision based only on applications could overstate economic strength if survival rates fall. The ideal next step would link applications to realized employer births, payroll records, or transaction data. The public sensitivity index can guide where to look, but it should not be treated as the final outcome of interest.

The broader lesson is that measurement tools should be built around the decision they inform. If the question is where to allocate small-business support, the relevant measure may combine applications, retail demand, credit conditions, and labor-market distress. If the question is where new digital businesses are likely to emerge, the relevant measure may weight high-propensity applications more heavily and include online-sector proxies. If the question is macroeconomic nowcasting, the relevant measure may prioritize forecast accuracy. This article provides one transparent building block: the sensitivity of state business formation to local retail demand.

8 Additional State-Level Evidence

The sensitivity index can be used to classify states into broad regimes. High-sensitivity states are places where application growth tends to rise when retail demand rises. Low-sensitivity states are places where applications appear to be driven by other factors or where retail demand is a weak proxy for local opportunity. This classification is not an evaluation of policy quality. A low-sensitivity state may have entrepreneurship concentrated in nonretail sectors. A high-sensitivity state may have more consumer-facing entry. The index is useful because it organizes heterogeneity that would otherwise be hidden by national averages.

The result also warns against interpreting national business-application numbers without local context. A national surge can arise from broad digital opportunities, from a few large states, from many small state-specific increases, or from sectoral changes that are not tied to local retail demand. State panels allow researchers to ask whether national movements are geographically broad and whether local demand proxies explain them. In this paper, the positive baseline association and the interaction-period attenuation indicate that the entry margin is partly local and partly driven by broader regime-specific forces.

Because the data are monthly, timing matters. An entrepreneur may respond to demand with a lag if she needs to arrange financing, secure suppliers, or observe whether a demand increase persists. Conversely, applications may lead realized retail sales if entrepreneurs anticipate demand. The baseline fixed-effects specification uses contemporaneous retail growth because the objective is an interpretable sensitivity index. The state-specific ranking includes one lag. Future work could estimate distributed-lag models or local projections, but the short panel and the volatility of both series caution against overly flexible dynamics.

The public-data workflow is designed to be updated. Each new month of BFS and MSRS data can be appended, the panel can be regenerated, and the sensitivity index can be revised. That makes the measure potentially useful for ongoing monitoring. At the same time, revised data can change the historical series. An applied researcher should therefore distinguish between final-vintage analysis and real-time monitoring. The code records the raw vintage used for the article and can be adapted to store future vintages.

9 Conclusion

This paper links public Business Formation Statistics to Monthly State Retail Sales data to measure the state-level sensitivity of high-propensity business applications to local retail demand. The baseline two-way fixed-effects estimate is positive: a one percentage point increase in retail growth is associated with 0.48 percentage points higher high-propensity application growth, with a clustered standard error of 0.28. The association is attenuated during the pandemic and after March 2022, indicating that the demand-entry relationship is regime-dependent. State-specific sensitivity estimates reveal substantial heterogeneity across states.

The paper's contribution is a reproducible measurement exercise rather than a causal elasticity. It shows how messy public data can be converted into a state-month panel and a useful economic index. The limitations are equally clear: retail growth is endogenous, the state retail data are experimental, and applications are not firm births. The appropriate conclusion is therefore bounded. Public data can identify where business formation comoves with local demand, but stronger causal and administrative data are needed to determine why and whether those applications become durable firms.

10 Data and Code Availability

The empirical results are generated from public Census Business Formation Statistics and Monthly State Retail Sales files. The analysis code parses the raw sources, reshapes the state-month business-application panel, merges retail-growth measures, estimates the fixed-effects specifications, constructs the state sensitivity index, and writes the figures, tables, and LaTeX macros used in the article. The paper therefore does not rely on hand-entered estimates.

The code records intermediate outputs, including the state-month panel, period indicators, residualized variables, and state-specific sensitivity estimates. These files make the empirical design auditable and allow readers to evaluate alternative windows, transformations, or state-ranking procedures without relying on proprietary data.

The analysis uses public data files available in June 2026. Because Census series are revised and extended over time, exact values may change when the code is rerun against later releases. The raw-file manifest records hashes for the vintage used here so that future reruns can distinguish data revisions from code changes.

A Additional Empirical Details and Alternative Specifications

This appendix discusses alternative designs for the small-business sensitivity index. The baseline paper estimates a descriptive state-month relationship between retail growth and high-propensity business applications. It does not claim that retail growth is as-good-as-random. This distinction should remain central to the interpretation. A causal version of the paper would need a source of local demand variation that affects expected revenue but is otherwise unrelated to entry. Examples might include predetermined industry exposure to national demand shocks, exogenous weather shocks to local retail activity, or policy discontinuities that shift consumer spending without directly changing business-formation incentives.

A natural extension is a shift-share design. One could measure each state's pre-period retail industry composition and interact it with national industry-specific retail shocks excluding the state itself. The resulting predicted retail-demand shock would be less endogenous to state-specific entrepreneurial conditions than observed state retail growth. Such a design would still require careful treatment of exposure exogeneity and industry-level shocks, but it would move the paper closer to causal interpretation. The present article does not implement this extension because the state retail file used here is already an aggregate year-over-year panel and the baseline design is meant to remain transparent.

Another extension is a distributed-lag model. Business applications may respond to retail demand with delays. A potential entrepreneur might observe several months of stronger demand before filing. Alternatively, applications may lead retail sales if entrepreneurs anticipate future demand. The baseline contemporaneous specification is easy to interpret but cannot distinguish these timing stories. The state-specific sensitivity index includes a one-month lag as a simple diagnostic. A richer version would estimate lag structures by period and test whether the pandemic altered the timing of the relationship.

The interaction periods could also be refined. The baseline uses a pandemic indicator and a post-March-2022 indicator. These are coarse windows that approximate economic regimes rather than exact structural breaks. A formal approach could estimate rolling coefficients, use time-varying-parameter models, or test candidate break dates. The difficulty is that the panel is short and the shock periods are unusual. More flexible time variation may overfit. The baseline therefore uses broad, economically motivated periods. The interpretation is not that March 2022 is a precise break date; it is that the demand-entry relationship appears weaker in the later inflation and tightening period.

State heterogeneity is central to the paper. The state-specific sensitivity estimates reveal variation that is hidden by pooled regressions. However, those estimates are noisy because each state has a limited monthly time series and because both retail growth and applications are volatile. A future version could improve this by using hierarchical shrinkage, empirical Bayes estimates, or random-coefficient models. Those methods would produce more stable state rankings by borrowing strength across states. The tradeoff is that the resulting index would be less transparent. The current paper reports simple state regressions because they are easy to audit.

The choice of high-propensity applications matters. All applications include many filings that are unlikely to become employer firms. High-propensity applications are closer to potential employer entry, but they can still include businesses that never hire or survive. A more complete analysis would link applications to subsequent employer births and payroll using longitudinal administrative data. Public BFS does not provide that linkage at the individual application level. The article therefore studies the upstream margin. This is appropriate for a real-time indicator, but it should not be confused with realized small-business growth.

The Monthly State Retail Sales series is also imperfect. It is an experimental product and may be revised. Retail growth is only one component of local demand. Some states have large nonretail entrepreneurial sectors, such as professional services, construction, energy, tourism, technology, or health care. In those states, total retail growth may be weakly related to entry opportunities. This compositional issue may explain why some state sensitivities are low or negative. A sector-specific paper would match business applications by industry to retail or services demand by industry, but that would require a more detailed data build.

A concern with two-way fixed effects is that they remove national shocks but not spatial spillovers. Entrepreneurs can start businesses serving customers outside their state, especially online. Retail demand in one state can also spill over to neighboring states through commuting, tourism, and cross-border shopping. If applications respond to national or regional demand rather than state demand, the within-state retail coefficient will understate broader demand sensitivity. This issue is not a flaw in the regression; it defines what the coefficient measures. The paper measures state-local comovement, not national digital opportunity.

The binned scatter is included because tables alone can hide the identifying variation. After residualizing state and month effects, the relationship between retail growth and application growth is positive but dispersed. This visual evidence is important for readers. It shows that the baseline coefficient is not driven only by time-series levels or by cross-state differences, but it also shows substantial noise. In applied economics, such figures are often more informative than a long list of robustness coefficients because they display the empirical support for the main slope.

One could also construct a sensitivity index at a metropolitan or county level if the required data were available. That would be preferable for many economic questions because business formation and consumer demand are local. State averages are coarse and mix very different local economies.

However, the public MSRS data are state-level, and the BFS state series are the most readily available high-frequency application outcomes. The state design is therefore a compromise between frequency, coverage, and geographic detail. A firm or agency with administrative transaction data could construct much more granular measures.

The paper's results should not be interpreted as evidence that small businesses are unimportant. The analysis studies the sensitivity of applications to retail demand, not the level contribution of small firms to employment or output. A state can have high entrepreneurial dynamism but low retail-demand sensitivity if entry is driven by technology, professional services, or migration. Conversely, a state can have high sensitivity because entry is concentrated in consumer-facing sectors. The index is a diagnostic of one channel, not a ranking of economic quality.

The panel could be augmented with macro-financial variables. Interest rates are national and are absorbed by month fixed effects, but their local impact may vary with credit dependence, housing wealth, or industry mix. Inflation may also matter differently across states. A future version could interact national financial conditions with predetermined state exposure measures. That would turn the sensitivity index into a broader small-business stress model. The current paper focuses on retail demand because it is the cleanest public state-month variable that maps to potential customer demand.

Finally, the article's contribution lies in the full empirical workflow. It begins with a business-relevant question, turns it into an estimable economic object, builds a public data panel, estimates interpretable models, reports limitations, and documents the data construction. The same logic could be applied inside a data-rich institution by replacing the public retail proxy and applications with internal transaction and merchant data. The public analysis demonstrates the research logic without relying on confidential information.

The core conclusion is therefore bounded but useful. High-propensity business applications co-move positively with retail demand in baseline state-month variation, but the relationship weakens during major macro regimes and varies across states. A sensitivity index summarizes that relationship in a way that can be updated and audited. Stronger causal designs and richer administrative data are needed to explain the heterogeneity and to connect applications to durable firm outcomes.

B Additional Notes on Business-Formation Measurement

The sensitivity index is also useful because it forces the analyst to confront the difference between entry intentions and operating firms. Business applications are forward-looking administrative actions. They may rise when expected opportunities rise, but they can also rise when people experiment with side businesses, when tax or legal incentives change, or when labor-market disruptions encourage self-employment. This feature is not a flaw of the data. It is precisely why the series is informative before slower employer statistics are released. The correct empirical response is to use the series for what it measures: a timely upstream signal of potential entry. Any subsequent paper that links the sensitivity index to employment, payroll, revenue, or survival would add a second stage that is not available in the public files used here.

The distinction matters for interpreting cross-state heterogeneity. A high-sensitivity state may have an application margin that responds quickly to consumer-facing demand, but it may not have a high conversion rate from applications to durable employer firms. A low-sensitivity state may have applications driven by professional services, construction, or technology rather than retail demand. Thus, the sensitivity ranking should not be used as a policy scoreboard. It is an empirical map of one relationship. The value of the map is that it tells researchers where to look more carefully and what kinds of additional data would be needed to explain the patterns.

A final detail concerns communication. The central coefficient is simple enough to explain without losing the economics: after removing permanent state differences and national monthly shocks, states with stronger retail growth tend to have stronger high-propensity application growth, but that relationship changes in unusual macro regimes. That sentence is the object the tables estimate. The surrounding article then explains why the coefficient is not causal, why the relationship varies over time, and why richer data are needed. This layered communication is useful for journal readers and for applied research teams alike.

The preferred next empirical step is therefore clear. Keep the sensitivity index as the descriptive benchmark, then add an identification design that predicts retail demand using predetermined exposure. If the causal estimate and the descriptive sensitivity point in the same direction, the evidence becomes stronger. If they differ, the difference itself would reveal how much of the public comovement reflects common shocks rather than demand-induced entry. That comparison would sharpen the article's causal interpretation while preserving the descriptive index.

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